

Aarsha Anil

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EDUCATION

- Vimal Jyothi Engineering College** Kannur, Kerala
B.Tech in AI & Data Science; GPA: 9.08 2020 – 2024

EXPERIENCE

- AI Intern** Sep 2024 – Dec 2024
Innomatics Research Labs Remote, India
 - LLM Fine-tuning & LangChain Integration:** Implemented fine-tuning workflows on transformer models using LangChain framework built custom intelligent agents with RAG, memory, and tool-calling capabilities.
 - Retrieval-Augmented Generation (RAG) Pipelines:** Developed RAG pipelines integrating vector databases (e.g., FAISS) with OpenAI APIs for contextual question answering on large clinical datasets.
 - Prompt Engineering & Model Optimization:** Designed prompt templates and optimized workflows to improve LLM output relevance and efficiency for domain-specific applications.
 - Deployment:** Deployed AI microservices with Docker and AWS Lambda and implemented CI/CD pipelines with GitHub Actions, improving reliability and monitoring.
- Data Science Intern** Feb 2024 – Apr 2024
Coapps Development Solutions India
 - Deep Learning Model Training:** Fine-tuned LSTM and GRU models for sales forecasting on Azure ML, improving accuracy by 12%.
 - Fraud Detection Using Explainable ML:** Built interpretable fraud detection pipelines using PyTorch.
 - Dashboard Development:** Created interactive Power BI dashboards for image analytics and monitoring key product metrics.

PROJECTS

- Distributed Multi-Agent GenAI System:** Built a decentralized multi-agent system orchestrating Gemini and LLaMA 3 models via CrewAI and LangGraph; implemented dynamic workflows with LangChain for market research and media analytics.
- Vision-Based Anomaly Detection:** Developed CNN models for real-time anomaly detection in surveillance videos, integrating OpenCV and PyTorch with alerting UI.
- YouTube-to-Blog Generator (NLP + LLM):** Automated transcript summarization and blog generation using LangChain agents, with memory management and rate limiting for efficient pipeline execution.
- Twitter Analytics – Real-Time BI:** Built Power BI dashboards with Python-based sentiment analysis on live Twitter streams using Tweepy and NLP models.

TECHNICAL SKILLS

- Languages:** Python, SQL, R, Java, C, NoSQL **Frameworks & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, LangChain, LangGraph, OpenCV
- Tools:** Docker, FastAPI, Git, Kubernetes, GitHub Actions, Power BI **Platforms:** AWS (Lambda, EC2), Azure ML, GCP, Vector DBs (FAISS, Chroma)
- Concepts & Domains:** Large Language Models (LLMs), Fine-tuning, Prompt Engineering, RAG Pipelines, Deep Learning, Model Quantization, Supervised, Unsupervised learning